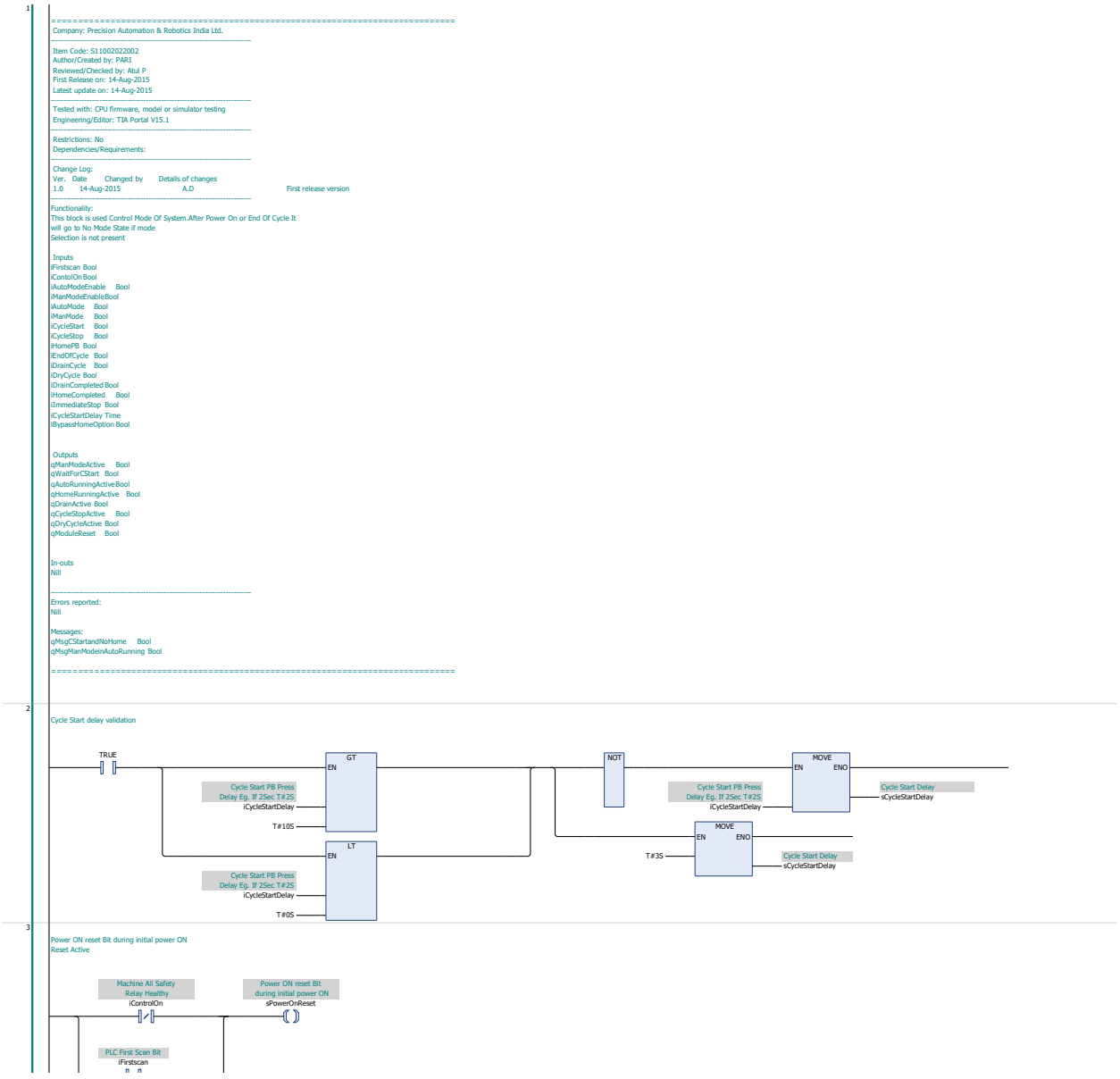
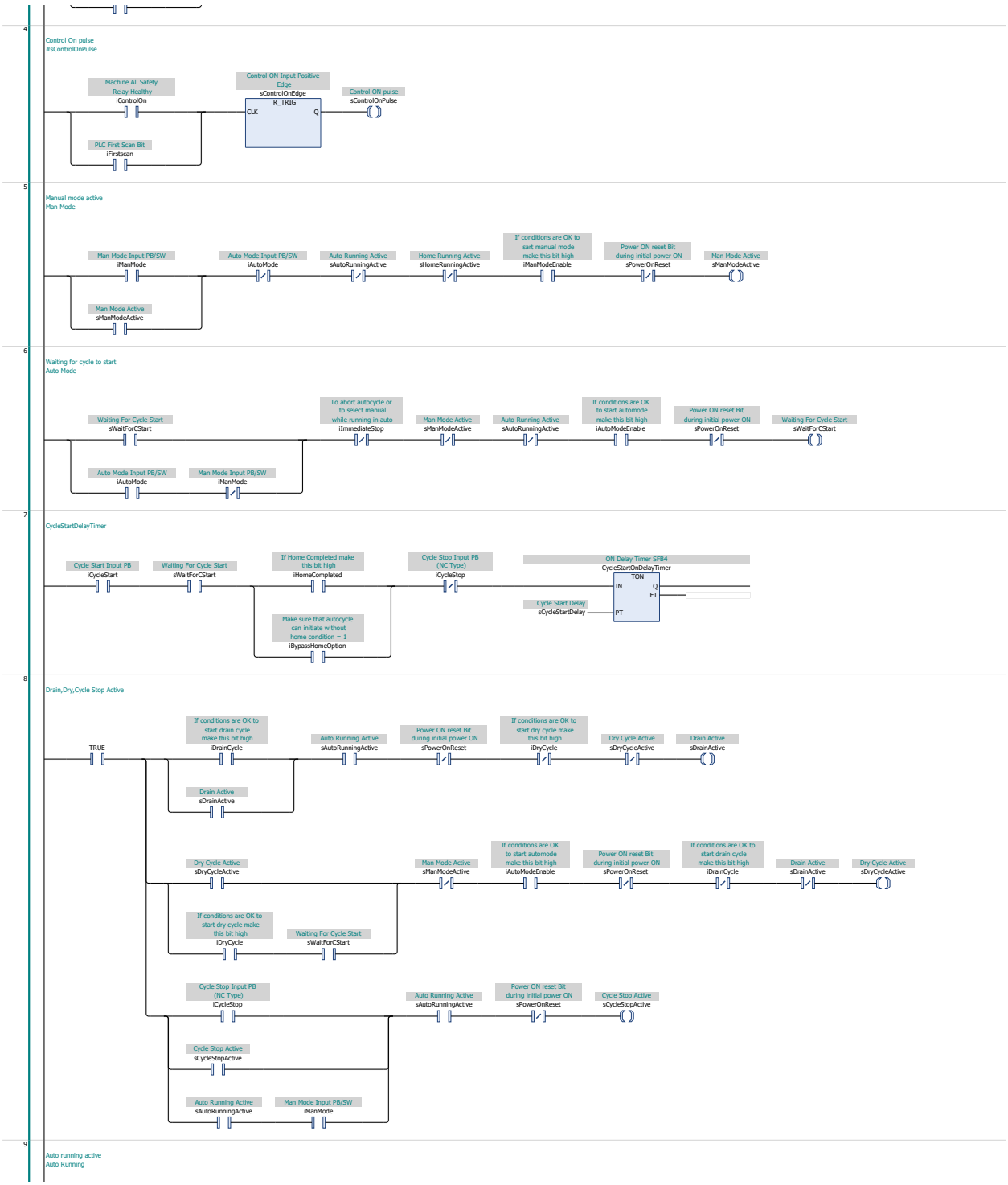
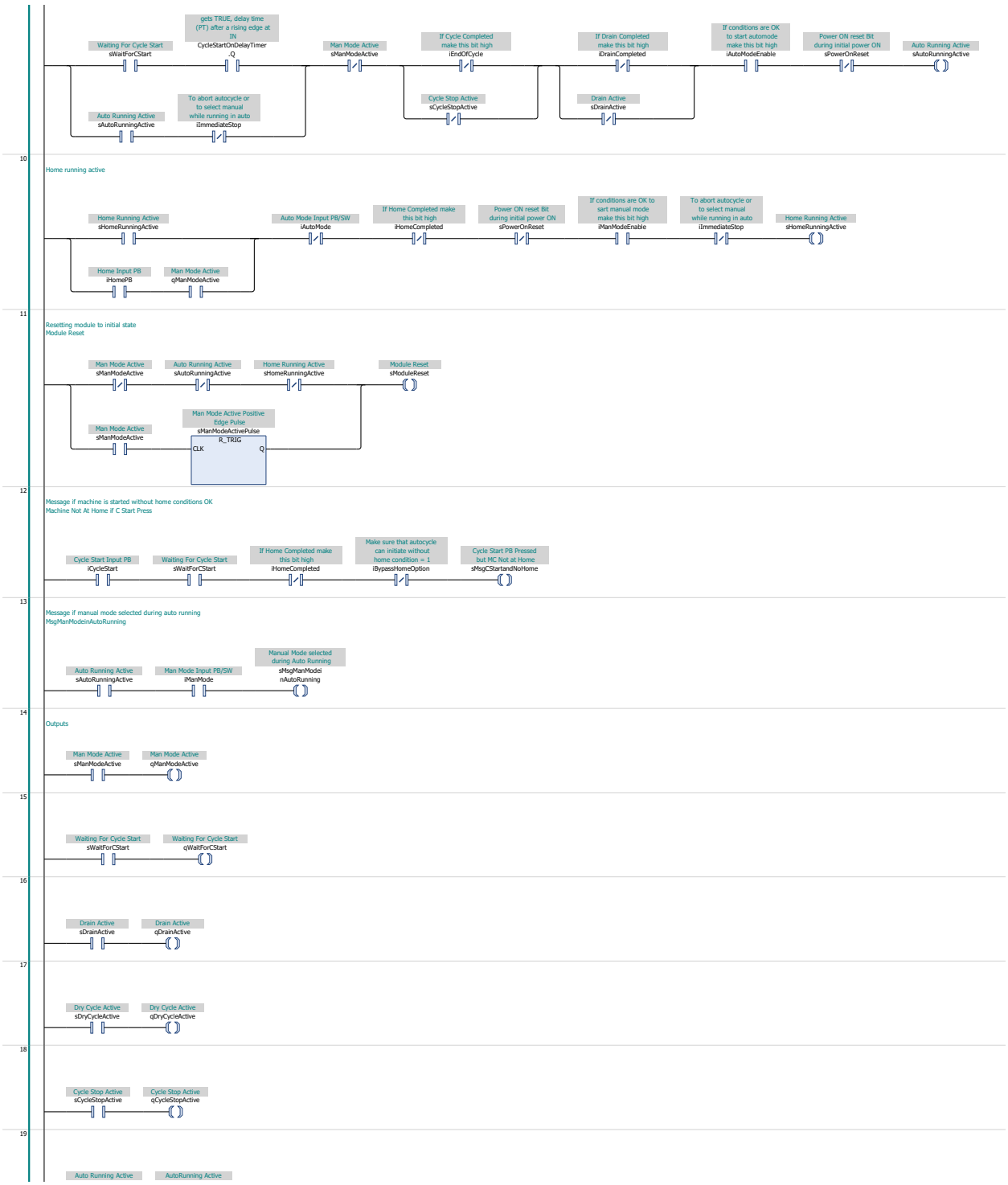
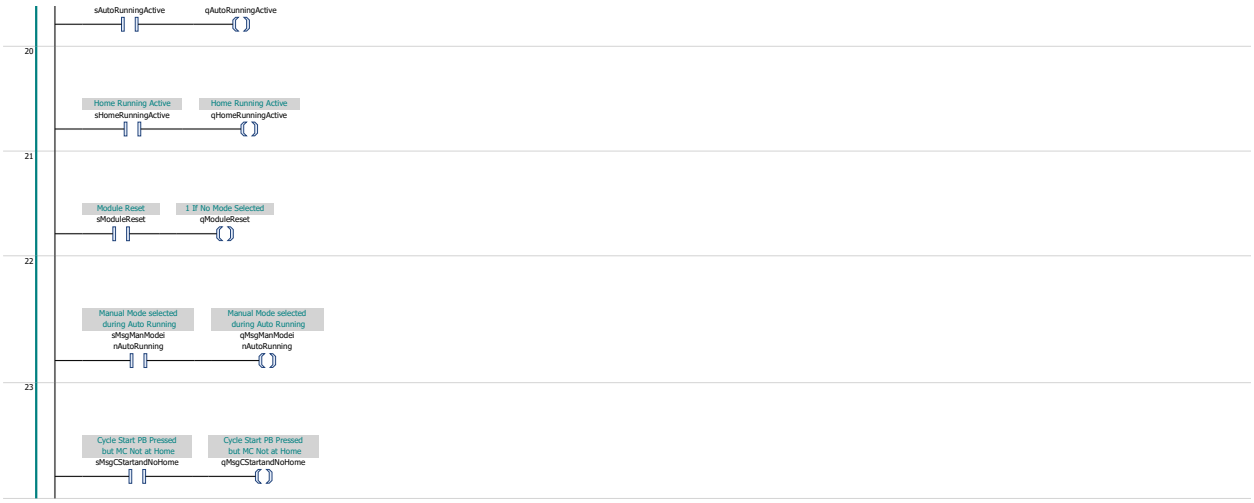
```
1   FUNCTION_BLOCK PR_ModeManageNoMode
2   VAR_INPUT
3       iFirstscan : BOOL ;  // PLC First Scan Bit
4       iControlOn : BOOL ;  // Machine All Safety Relay Healthy
5       iAutoModeEnable : BOOL ;  // If conditions are OK to start automode make
    this bit high
6       iManModeEnable : BOOL ;  // If conditions are OK to sart manual mode make
    this bit high
7       iAutoMode : BOOL ;  // Auto Mode Input PB/SW
8       iManMode : BOOL ;  // Man Mode Input PB/SW
9       iCycleStart : BOOL ;  // Cycle Start Input PB
10      iCycleStop : BOOL ;  // Cycle Stop Input PB (NC Type)
11      iHomePB : BOOL ;  // Home Input PB
12      iEndOfCycle : BOOL ;  // If Cycle Completed make this bit high
13      iDrainCycle : BOOL ;  // If conditions are OK to start drain cycle make
    this bit high
14      iDryCycle : BOOL ;  // If conditions are OK to start dry cycle make this
    bit high
15      iDrainCompleted : BOOL ;  // If Drain Completed make this bit high
16      iHomeCompleted : BOOL ;  // If Home Completed make this bit high
17      iImmediateStop : BOOL ;  // To abort autocycle or to select manual while
    running in auto make this bit 1
18      iCycleStartDelay : TIME ;  // Cycle Start PB Press Delay Eg. If 2Sec T#2S
19      iBypassHomeOption : BOOL ;  // Make sure that autocycle can initiate
    without home condition = 1
20  END_VAR
21  VAR_OUTPUT
22      qManModeActive : BOOL ;  // Man Mode Active
23      qWaitForCStart : BOOL ;  // Waiting For Cycle Start
24      qAutoRunningActive : BOOL ;  // AutoRunning Active
25      qHomeRunningActive : BOOL ;  // Home Running Active
26      qDrainActive : BOOL ;  // Drain Active
27      qCycleStopActive : BOOL ;  // Cycle Stop Active
28      qDryCycleActive : BOOL ;  // Dry Cycle Active
29      qModuleReset : BOOL ;  // 1 If No Mode Selected
30      qMsgCStartandNoHome : BOOL ;  // Cycle Start PB Pressed but MC Not at Home
31      qMsgManModeinAutoRunning : BOOL ;  // Manual Mode selected during Auto
    Running
32  END_VAR
33  VAR
34      sPowerOnReset : BOOL ;  // Power ON reset Bit during initial power ON
35      sControlOnPulse : BOOL ;  // Control ON pulse
36      sControlOnEdge : R_TRIG ;  // Control ON Input Positive Edge
37      sManModeActivePulse : R_TRIG ;  // Man Mode Active Positive Edge Pulse
38      sCycleStartDelay : TIME ;  // Cycle Start Delay
39      CycleStartOnDelayTimer : TON ;  // ON Delay Timer SFB4
40      sAutoRunningActive : BOOL ;  // Auto Running Active
41      sManModeActive : BOOL ;  // Man Mode Active
42      sHomeRunningActive : BOOL ;  // Home Running Active
43      sWaitForCStart : BOOL ;  // Waiting For Cycle Start
44      sDrainActive : BOOL ;  // Drain Active
45      sDryCycleActive : BOOL ;  // Dry Cycle Active
```

```
46          sCycleStopActive  : BOOL ;   // Cycle Stop Active
47          sModuleReset  : BOOL ;   // Module Reset
48          sMsgCStartandNoHome  : BOOL ;   // Cycle Start PB Pressed but MC Not at Home
49          sMsgManModeinAutoRunning  : BOOL ;   // Manual Mode selected during Auto
        Running
50      END_VAR
51
```




4
Control On pulse
#sControlOnPulse
Machine All Safety Relay Healthy
iControlOn
Control ON Input Positive Edge
sControlOnEdge
R_TRIG
CLK
Q
Control ON pulse
sControlOnPulse
PLC First Scan Bit
iFirstscan
5
Manual mode active
Man Mode
Man Mode Input PB/SW
iManMode
Auto Mode Input PB/SW
iAutoMode
Auto Running Active
sAutoRunningActive
Home Running Active
sHomeRunningActive
If conditions are OK to sart manual mode make this bit high
iManModeEnable
Power ON reset Bit during initial power ON
sPowerOnReset
Man Mode Active
sManModeActive
Man Mode Active
sManModeActive
6
Waiting for cycle to start
Auto Mode
Waiting For Cycle Start
sWaitForCStart
To abort autocycle or to select manual while running in auto
iImmediateStop
Man Mode Active
sManModeActive
Auto Running Active
sAutoRunningActive
If conditions are OK to start automode make this bit high
iAutoModeEnable
Power ON reset Bit during initial power ON
sPowerOnReset
Waiting For Cycle Start
sWaitForCStart
Auto Mode Input PB/SW
iAutoMode
Man Mode Input PB/SW
iManMode
7
CycleStartDelayTimer
Cycle Start Input PB
iCycleStart
Waiting For Cycle Start
sWaitForCStart
If Home Completed make this bit high
iHomeCompleted
Cycle Stop Input PB (NC Type)
iCycleStop
ON Delay Timer SFB4
CycleStartOnDelayTimer
TON
IN
Q
ET
PT
Cycle Start Delay
sCycleStartDelay
Make sure that autocycle can initiate without home condition = 1
iBypassHomeOption
8
Drain,Dry,Cycle Stop Active
TRUE
If conditions are OK to start drain cycle make this bit high
iDrainCycle
Auto Running Active
sAutoRunningActive
Power ON reset Bit during initial power ON
sPowerOnReset
If conditions are OK to start dry cycle make this bit high
iDryCycle
Dry Cycle Active
sDryCycleActive
Drain Active
sDrainActive
Drain Active
sDrainActive
Dry Cycle Active
sDryCycleActive
Man Mode Active
sManModeActive
If conditions are OK to start automode make this bit high
iAutoModeEnable
Power ON reset Bit during initial power ON
sPowerOnReset
If conditions are OK to start drain cycle make this bit high
iDrainCycle
Drain Active
sDrainActive
Dry Cycle Active
sDryCycleActive
If conditions are OK to start dry cycle make this bit high
iDryCycle
Waiting For Cycle Start
sWaitForCStart
Cycle Stop Input PB (NC Type)
iCycleStop
Auto Running Active
sAutoRunningActive
Power ON reset Bit during initial power ON
sPowerOnReset
Cycle Stop Active
sCycleStopActive
Cycle Stop Active
sCycleStopActive
Auto Running Active
sAutoRunningActive
Man Mode Input PB/SW
iManMode
9
Auto running active
Auto Running


gets TRUE, delay time (PT) after a rising edge at IN
Waiting For Cycle Start
sWaitForCStart
CycleStartOnDelayTimer
.Q
Man Mode Active
sManModeActive
If Cycle Completed make this bit high
iEndOfCycle
If Drain Completed make this bit high
iDrainCompleted
If conditions are OK to start automode make this bit high
iAutoModeEnable
Power ON reset Bit during initial power ON
sPowerOnReset
Auto Running Active
sAutoRunningActive
To abort autocycle or to select manual while running in auto
iImmediateStop
Cycle Stop Active
sCycleStopActive
Drain Active
sDrainActive
10
Home running active
Home Running Active
sHomeRunningActive
Auto Mode Input PB/SW
iAutoMode
If Home Completed make this bit high
iHomeCompleted
Power ON reset Bit during initial power ON
sPowerOnReset
If conditions are OK to sart manual mode make this bit high
iManModeEnable
To abort autocycle or to select manual while running in auto
iImmediateStop
Home Running Active
sHomeRunningActive
Home Input PB
iHomePB
Man Mode Active
qManModeActive
11
Resetting module to initial state
Module Reset
Man Mode Active
sManModeActive
Auto Running Active
sAutoRunningActive
Home Running Active
sHomeRunningActive
Module Reset
sModuleReset
Man Mode Active Positive Edge Pulse
sManModeActivePulse
R_TRIG
CLK
Q
12
Message if machine is started without home conditions OK
Machine Not At Home if C Start Press
Cycle Start Input PB
iCycleStart
Waiting For Cycle Start
sWaitForCStart
If Home Completed make this bit high
iHomeCompleted
Make sure that autocycle can initiate without home condition = 1
iBypassHomeOption
Cycle Start PB Pressed but MC Not at Home
sMsgCStartandNoHome
13
Message if manual mode selected during auto running
MsgManModeinAutoRunning
Auto Running Active
sAutoRunningActive
Man Mode Input PB/SW
iManMode
Manual Mode selected during Auto Running
sMsgManModeinAutoRunning
14
Outputs
Man Mode Active
sManModeActive
Man Mode Active
qManModeActive
15
Waiting For Cycle Start
sWaitForCStart
Waiting For Cycle Start
qWaitForCStart
16
Drain Active
sDrainActive
Drain Active
qDrainActive
17
Dry Cycle Active
sDryCycleActive
Dry Cycle Active
qDryCycleActive
18
Cycle Stop Active
sCycleStopActive
Cycle Stop Active
qCycleStopActive
19
Auto Running Active
AutoRunning Active


sAutoRunningActive
qAutoRunningActive
20
Home Running Active
sHomeRunningActive
Home Running Active
qHomeRunningActive
21
Module Reset
sModuleReset
1 If No Mode Selected
qModuleReset
22
Manual Mode selected during Auto Running
sMsgManModei
nAutoRunning
Manual Mode selected during Auto Running
qMsgManModei
nAutoRunning
23
Cycle Start PB Pressed but MC Not at Home
sMsgCStartandNoHome
Cycle Start PB Pressed but MC Not at Home
qMsgCStartandNoHome